



Course Specification

(Bachelor)

Course Title: Senior Design Project
Course Code: EE 490
Program: Electrical Engineering
Department: Communications and Networks Engineering
College: College of Engineering
Institution: Prince Sultan University
Version: 1
Last Revision Date: 25-09-2024

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A. General information about the course:

1. Course Identification

1. Credit hours: (3)				
3				
2. Course type: Capstone				
A.	☐ University	☐ College	☒ Program	☐ Track
B.	☒ Required		☐ Elective	
3. Level/year at which this course is offered: (7 / 4)				
4. Course general Description:				
Project groups deliver their products that have progressed through the design, analysis, testing and evaluation stages. The project teams produce a polished professional report that describes the design process, implementation and testing, verification and validation, and a critical appraisal of the project. An oral presentation and graphic works are complementary project deliverables				
5. Pre-requirements for this course (if any):				
ENG 301, CEE 201, Completion of 90 Credit Hours				
6. Co-requisites for this course (if any):				
ETHC305				

7. Course Main Objective(s):				
• Apply engineering knowledge and design methodology to develop a functional solution to a real-world electrical engineering problem. • Demonstrate effective teamwork, project management, and professional communication skills through all phases of the design process. • Evaluate design outcomes with consideration for ethical, environmental, societal, and economic factors, and present the final product through comprehensive technical documentation and oral presentations.				

2. Teaching mode (mark all that apply)

No	Mode of Instruction	Contact Hours	Percentage
1	Traditional classroom	45	100 %
2	E-learning		





No	Mode of Instruction	Contact Hours	Percentage
3	Hybrid - Traditional classroom - E-learning		
4	Distance learning		

3. Contact Hours (based on the academic semester)

No	Activity	Contact Hours
1.	Lectures	45
2.	Laboratory/Studio	
3.	Field	
4.	Tutorial	
5.	Others (specify)	
Total		45

B. Course Learning Outcomes (CLOs), Teaching Strategies and Assessment Methods

Code	Course Learning Outcomes	Code of CLOs aligned with program	Teaching Strategies	Assessment Methods
1.0	Knowledge and understanding			
1.1	Identify relevant technical information and engineering principles needed to define and solve an open-ended design problem.	1	Flipped Classroom; Project Studio; Laboratory Demonstrations ; Oral Discussions; Team Meetings	Written Reports (Proposal, Interim, and Final); Oral Presentations; Weekly Progress Assessment
2.0	Skills			
2.1	Apply the engineering design process to produce solutions that meet specified needs with consideration of safety,	2	Flipped Classroom; Project Studio; Laboratory Demonstrations ; Oral Discussions;	Written Reports (Proposal, Interim, and Final); Oral Presentations; Weekly Progress Assessment





Code	Course Learning Outcomes	Code of CLOs aligned with program	Teaching Strategies	Assessment Methods
	environmental, economic, and societal constraints and relevant standards/codes.		Team Meetings	
2.2	Generate alternative concepts and select a preferred solution for a design problem.	2	Flipped Classroom; Project Studio; Laboratory Demonstrations ; Oral Discussions; Team Meetings	Written Reports (Proposal, Interim, and Final); Oral Presentations; Weekly Progress Assessment
2.3	Communicate design solutions visually using engineering tools and drawings.	3	Flipped Classroom; Project Studio; Laboratory Demonstrations ; Oral Discussions; Team Meetings	Written Reports (Proposal, Interim, and Final); Oral Presentations; Weekly Progress Assessment
2.4	Communicate design solutions using written, visual, and oral technical methods.	3	Flipped Classroom; Project Studio; Laboratory Demonstrations ; Oral Discussions; Team Meetings	Written Reports (Proposal, Interim, and Final); Oral Presentations; Weekly Progress Assessment
2.5	Validate design solutions through experimentation/testing, data collection, analysis, and engineering conclusions.	6	Flipped Classroom; Project Studio; Laboratory Demonstrations ; Oral Discussions; Team Meetings	Written Reports (Proposal, Interim, and Final); Oral Presentations; Weekly Progress Assessment
3.0	Values, autonomy, and responsibility			
3.1	Evaluate ethical and legal responsibilities	4	Flipped Classroom;	Written Reports (Proposal, Interim,





Code	Course Learning Outcomes	Code of CLOs aligned with program	Teaching Strategies	Assessment Methods
	and their impact on engineering decisions.		Project Studio; Laboratory Demonstrations ; Oral Discussions; Team Meetings	and Final); Oral Presentations; Weekly Progress Assessment
3.2	Collaborate to establish goals, plan tasks, and meet project objectives within a team.	5	Flipped Classroom; Project Studio; Laboratory Demonstrations ; Oral Discussions; Team Meetings	Written Reports (Proposal, Interim, and Final); Oral Presentations; Weekly Progress Assessment
3.3	Acquire new knowledge, tools, standards, or technologies as needed to complete the design project.	7	Flipped Classroom; Project Studio; Laboratory Demonstrations ; Oral Discussions; Team Meetings	Written Reports (Proposal, Interim, and Final); Oral Presentations; Weekly Progress Assessment

C. Course Content

No	List of Topics	Contact Hours
1.		
Total		

D. Students Assessment Activities

Courses with no Laboratories (3 Credit hours) follow the following table:

No	Assessment Activities *	Assessment timing (in week no)	Percentage of Total Assessment Score
1.	Attendance and participation	All weeks	5%
2.	Homework Assignment	All weeks	5%
3.	Quizzes	4, 8	10%
4.	Major Exams	7, 11	40%



No	Assessment Activities *	Assessment timing (in week no)	Percentage of Total Assessment Score
5.	Final Exam	16 - 17	40%

Courses with Laboratories (4 Credit hours) follow the following table:

No	Assessment Activities *	Assessment timing (in week no)	Percentage of Total Assessment Score
1.	Attendance and participation	All weeks	5%
2.	Homework Assignment	All weeks	5%
3.	Quizzes	4, 8	5%
4.	Laboratory	All weeks	15%
5.	Major Exams	7, 11	30%
6.	Final Exam	16 - 17	40%

*Assessment Activities (i.e., Written test, oral test, oral presentation, group project, essay, etc.).

E. Learning Resources and Facilities

1. References and Learning Resources

Essential References	Exploring Engineering: An Introduction to Engineering and Design / Philip Kosky, ISBN 978-0-12-374723-5
Supportive References	
Electronic Materials	PSU EduHub, PSU LMS: https://lms.psu.edu.sa/
Other Learning Materials	None

2. Required Facilities and equipment

Items	Resources
facilities (Classrooms, laboratories, exhibition rooms, simulation rooms, etc.)	Classroom (minimum capacity of 20 students)
Technology equipment (projector, smart board, software)	Smart board with internet access.
Other equipment (depending on the nature of the specialty)	

F. Assessment of Course Quality

Assessment Areas/Issues	Assessor	Assessment Methods
Effectiveness of teaching	Course coordinator,	Class observation,





Assessment Areas/Issues	Assessor	Assessment Methods
	Peers, Students	Faculty Evaluation, Peer Review, Course evaluation survey (Indirect)
Effectiveness of student assessment	Course Instructor	Course Report, Grade Report Analysis, Course Evaluation Survey (Indirect)
Quality of learning resources	Course Instructor, Students	Course report, Course Evaluation Survey (Indirect)
The extent to which CLOs have been achieved	Course Instructor, Students	CLO Assessment Analysis Course Exit Survey (Indirect), Course evaluation survey (Indirect)
Other		

Assessors (Students, Faculty, Program Leaders, Peer Reviewer, Others (specify))

Assessment Methods (Direct, Indirect)

G. Specification Approval

COUNCIL /COMMITTEE	DEPARTMENTAL COUNCIL
REFERENCE NO.	1
DATE	25-09-2024

